
CLINTON ENWEREM

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SUMMARY

Robotics Research Engineer and Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026. Research in dexterous manipulation, robot learning, motion planning, and feedback control, with Python/C++ implementations in simulation and on physical arm-hand systems.

SKILLS

Robotics: ROS/ROS2, ros2_control, MoveIt2, MuJoCo, Drake, Isaac Sim, Isaac Lab, Pinocchio, SocketCAN; **Motion Planning:** RRT*, Informed RRT*, Sampling-Based Planning, FCL Collision Checking; **Control:** Control Barrier Functions, Safety Filters, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Safe & Constrained Reinforcement Learning; **Robot Learning & Generative Models:** Imitation Learning & Behavior Cloning, ACT (Action Chunking Transformers), Dagger, Diffusion Policies (DP3), Flow-Matching Generative Models, Reinforcement Learning (PPO); **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Programming:** Python, C++, MATLAB, Bash; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi; **Dev Tools:** git, Docker; **Robots & EoATs:** Fairino FR3, xArm7, Unitree G1 (simulation), LEAP Hand v1, RealHand L6.

WORK EXPERIENCE

Graduate Research Assistant

Institute for Systems Research (ISR), University of Maryland

Aug. 2021 – Present
College Park, MD

- Develop grasp-synthesis, motion-planning, and feedback-control methods for Fairino FR3 and xArm7 arms with LEAP Hand v1 and RealHand L6 end effectors.
- Build synchronized tactile, proprioceptive, and visual data pipelines; train imitation and reinforcement-learning policies; and evaluate learned manipulation in Isaac Lab.

Research Intern

MATRIX Lab, USM at Southern Maryland

Jun. – Aug. 2022
California, MD

- Identified a discrete-time Crazyflie model from flight data and implemented a finite-horizon optimal formation controller in Python and Crazyswarm; executed leader-follower formation flights with 10 physical Crazyflie quadrotors under localization uncertainty.

Robotics Engineer

Kognitive Robotics

Mar. 2020 – Feb. 2021
Lagos, Nigeria

- Designed power and sensor-actuator interfaces in Altium, assembled a mobile robot, and implemented and tested its ROS software on physical hardware.

Robotics Trainee

Robotics & Artificial Intelligence Nigeria (RAIN)

Jan. 2020 – Feb. 2021
Ibadan, Nigeria

- Built ROS-based visual-SLAM, state-estimation, and flight-control software for an embedded quadrotor.

PROJECTS

ParcelStow — Dexterous Insertion Policy Evaluation (Paper, Code, Data & Models) | Isaac Lab, PyTorch, ACT

2026

- Built a Unitree G1 and RealHand L6 grasp-reorient-insert benchmark and trained three imitation-learning baselines on **297** expert demonstrations. The expert and ACT both reached **100%** nominal success; at 2× execution speed, success was **84%** for the expert and **53%** for ACT.

Grasp Distance Fields, Feedback Control for Grasp Execution (Project Page, arXiv) | C++, Python, QP, CBF-CLF

2026

- Implemented a CBF-CLF quadratic-program controller over a configuration-space grasp distance field. Physical experiments lifted **46 of 50** objects, retained a **median 94%** of the certified grasp margin, and solved the QP in **0.09 ms**.

realhand_l6_ros2, Dexterous-Hand ROS 2 Stack (Code, DOI) | C++, ROS 2, ros2_control, SocketCAN

2026

- Implemented a `hardware_interface::SystemInterface`, tactile-state export, contact-gated control, URDF/SRDF, MoveIt 2 configuration, mock hardware, virtual-CAN tests, and CI. Validated CAN communication, joint feedback, and tactile acquisition on a physical RealHand L6.

SELECTED PUBLICATIONS

- **C. Enwerem et al.** “Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert-Learner Comparison Across Task Execution Speeds.” *arXiv preprint*, 2026. [arXiv:2609.01453](https://arxiv.org/abs/2609.01453), [code](#), [data & models](#).
- **C. Enwerem et al.** “Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality.” *Preprint*, 2026. [arXiv:2608.00600](https://arxiv.org/abs/2608.00600), [project page](#).
- Two additional first-author papers will appear at IROS and CDC in 2026; five other first-author papers cover control, safe learning, and robotic manipulation.

EDUCATION

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present

B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0.

Aug. 2018